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KESTREL: Just Say Who's Missing

Shiva Shrestha and Honghui Xu

Abstract

Flying a drone through a search is hard, even for experienced pilots, and remote areas rarely have internet. KESTREL is a self-flying search-and-rescue drone. The operator types who is missing and where to look; a large language model on the drone turns the request into a flight plan, and colour and thermal cameras with on-board AI find the person. Everything runs on the drone, with automatic safety checks and a human GO before take-off.

Introduction

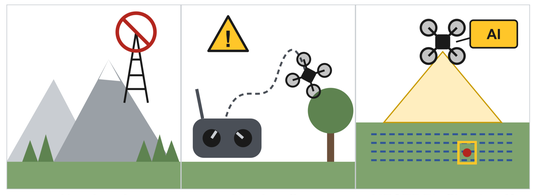
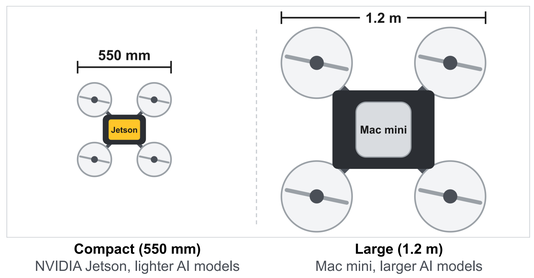


Fig. 1. Steering a drone through a search is hard even for skilled pilots, so KESTREL flies the search itself.

Research Questions

1. Can someone with no flying skills plan a search just by chatting with the drone?
2. Can all of the AI run on the drone itself, with no internet?
3. Can colour, heat and a written description together pick out one specific person?
4. Can an AI-planned flight stay safe and within FAA rules?

Materials: Two Drones



Compact (550 mm)
NVIDIA Jetson, lighter AI models

Large (1.2 m)
Mac mini, larger AI models

Fig. 2. The two KESTREL drones, drawn to scale (top view).

Materials and Methods

System overview

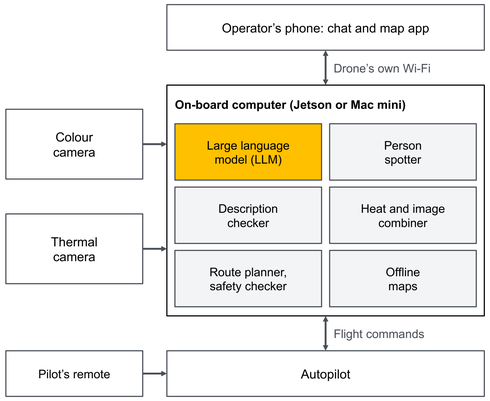


Fig. 3. How the parts connect. All the AI runs on the drone.

Planning a search by chatting

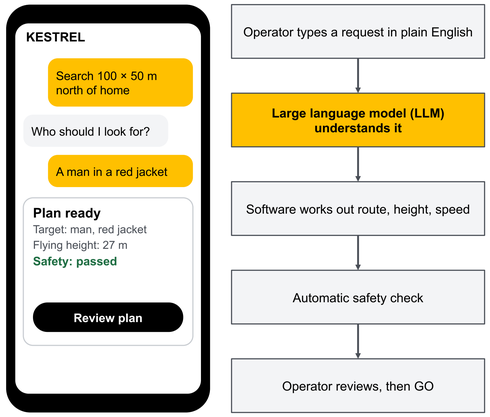


Fig. 4. The LLM only understands the words; regular software calculates every number.

Search pattern and flying height

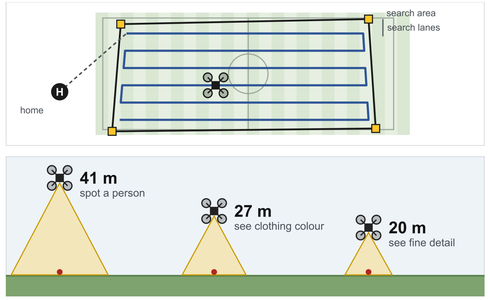
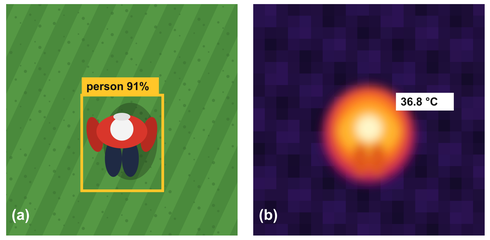


Fig. 5. Back-and-forth lanes cover the area; flying height is set by the detail needed.

Finding the person



1. Camera spots a person
2. Heat confirms a warm body
3. AI compares with the description

Match: photo and map pin sent to the phone

Fig. 6. (a) Colour and (b) thermal view of the same person (illustration).

Results

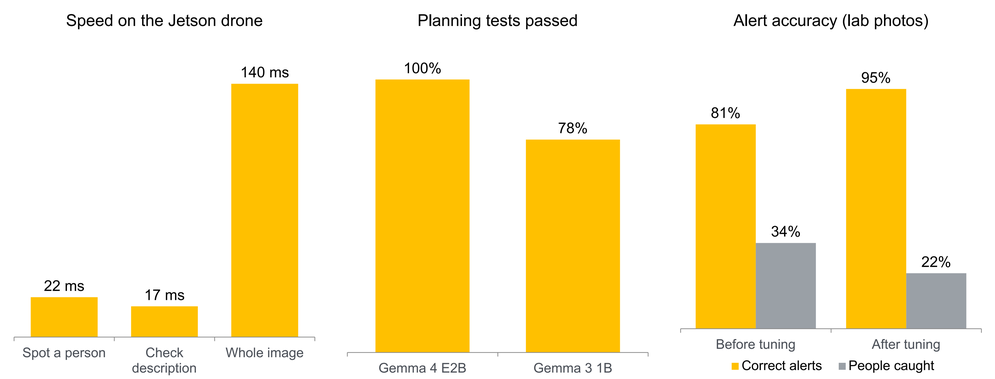


Fig. 7. Left: one image is fully analysed in 0.14 s. Middle: Gemma 4 E2B passed all 16 planning tests. Right: tuning cut false alarms but flags fewer matches.

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automatic software tests	internet needed	drone models

Safety and FAA Rules

- ✓ GPS locked
- ✓ Virtual fence
- ✓ Airspace approval
- ✓ Autopilot self-test
- ✓ Cameras and AI
- ✓ Battery level
- ✓ 400 ft limit
- ✓ Pilot's remote
- ✓ Route verified
- ✓ Legal checklist

Fig. 8. Checked automatically before every flight; any failure stops the launch.
FAA Part 107: a licensed pilot can take over at any time; the person running the search needs no flying skills.

Conclusions

- Anyone can plan a search by chatting; the LLM passed all 16 planning tests.
- All AI runs on the drone: one image takes 0.14 s on the Jetson model.
- Tuning raised alert accuracy to 95% on lab photos; real aerial tests are next.
- Ten automatic checks and two human approvals keep each flight safe.

Acknowledgments

Built with open-source software: ArduPilot, llama.cpp, Ultralytics YOLO, OpenCLIP, MapLibre and OpenStreetMap data.

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